



What is myside bias?

Definition, argument production [1]

- A tendency to produce arguments that...
- confirm one's own beliefs
 - dis-confirm views one rejects

Definition, general [2]

Evaluation & generation of evidence biased to favour own prior convictions

Negative effects from myside bias

- Belief polarisation [3]
- No agreement despite shared facts [4]
- No convergence to truth/best evidence [5]

Argumentation in humans tends to be myside biased

Data from	Mean <i>n</i> arguments favouring		Ratio
	my side	other side	
Perkins [6]	2.9	1.1	2.6
Toplak & Stanovich [7]	3.3	2.1	1.5

Question

The conclusions in myside biased arguments align with the arguer's prior beliefs. This bias is associated with negative epistemic outcomes.

But how does premise selection — the way in which we reason to support our prior beliefs — influence the epistemic effects of myside bias?

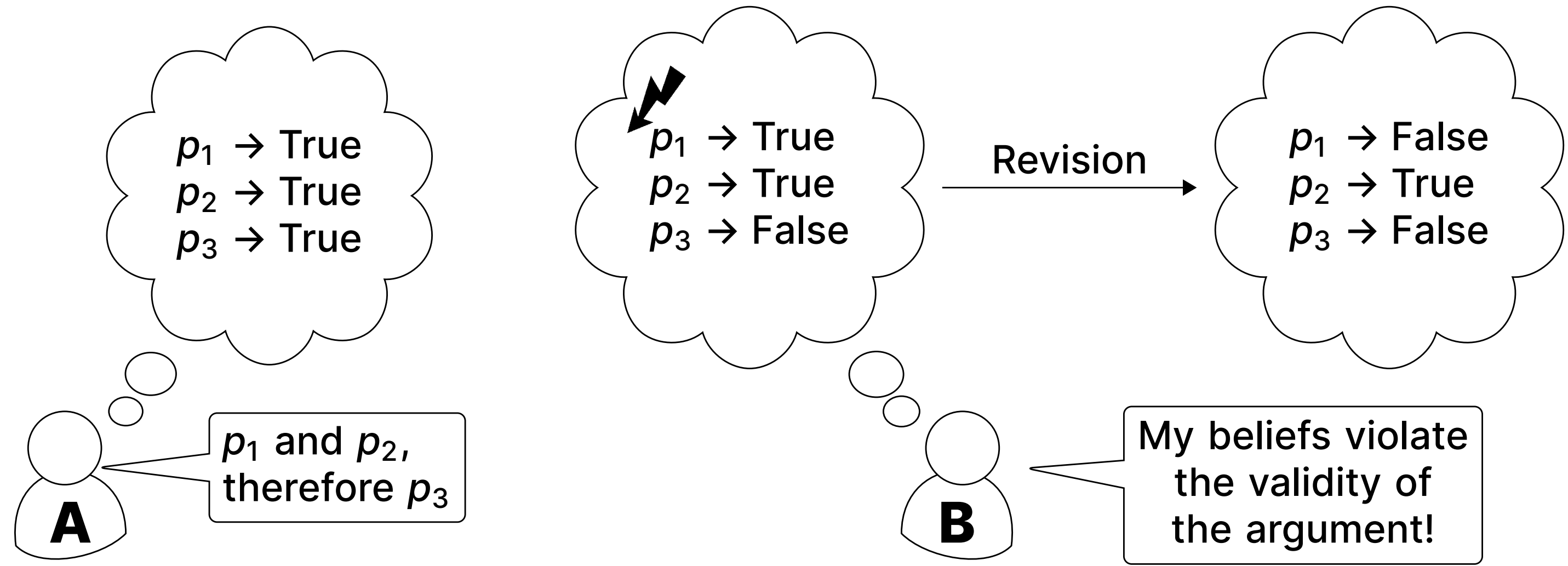
With a computational model we can study two different ways of selecting premises:

Option 1, egocentric: My beliefs support my prior belief

Option 2, common ground: Beliefs accepted by others support my prior belief

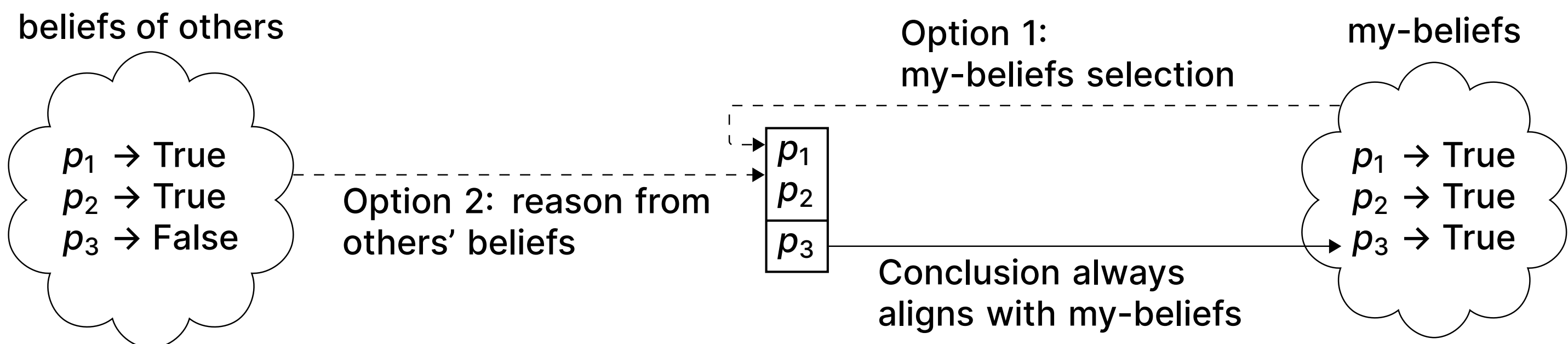
The model tracks the epistemic effects of these options in artificial agents.

A computational model of argumentation



Quality in, quality out: Agents only introduce valid arguments, respect the validity of arguments from others, and revise their beliefs accordingly.

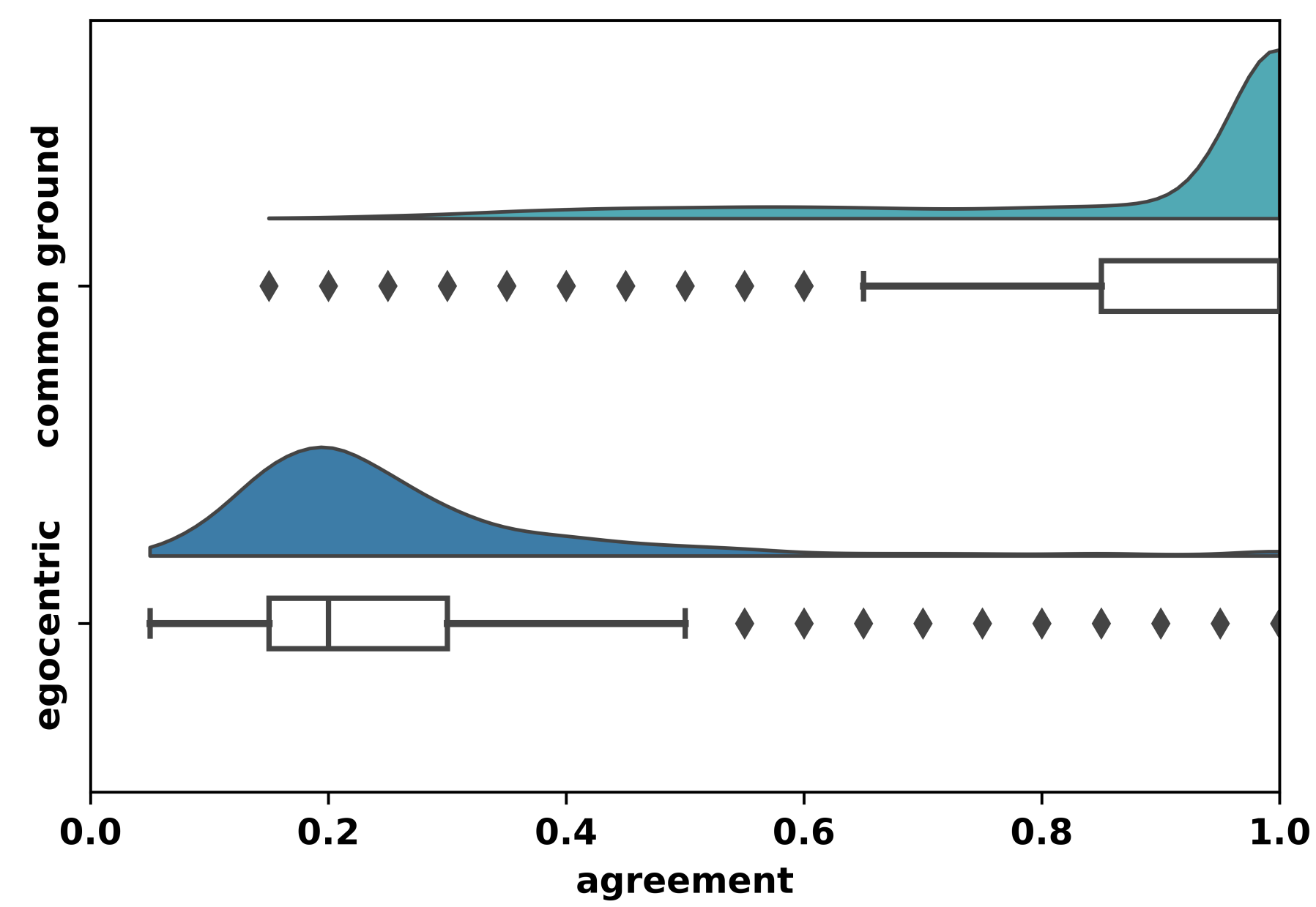
Argue for my-beliefs — but from which premises?



Simulation procedure

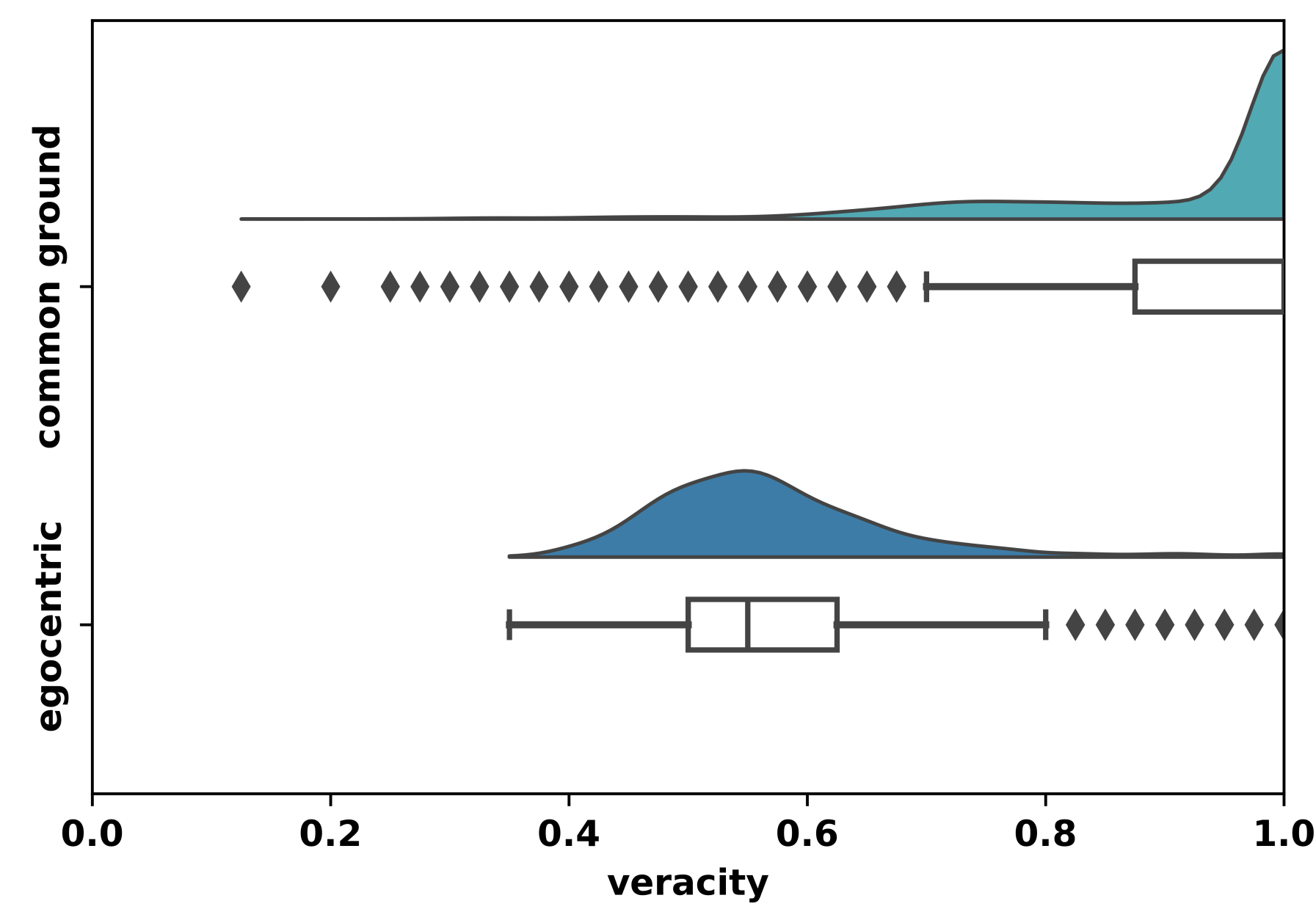
1. **Initialisation:** Agents hold random binary beliefs towards 20 propositions
2. **Introduction:** A randomly drawn agent introduces a valid argument
3. **Revision:** Agents that can't accommodate the validity of the new argument revise their beliefs, with minimal adaptation (repeat steps 2–3)
4. **Measurement:** Observe epistemic outcome at end of debate

Consensus building (2 agents)



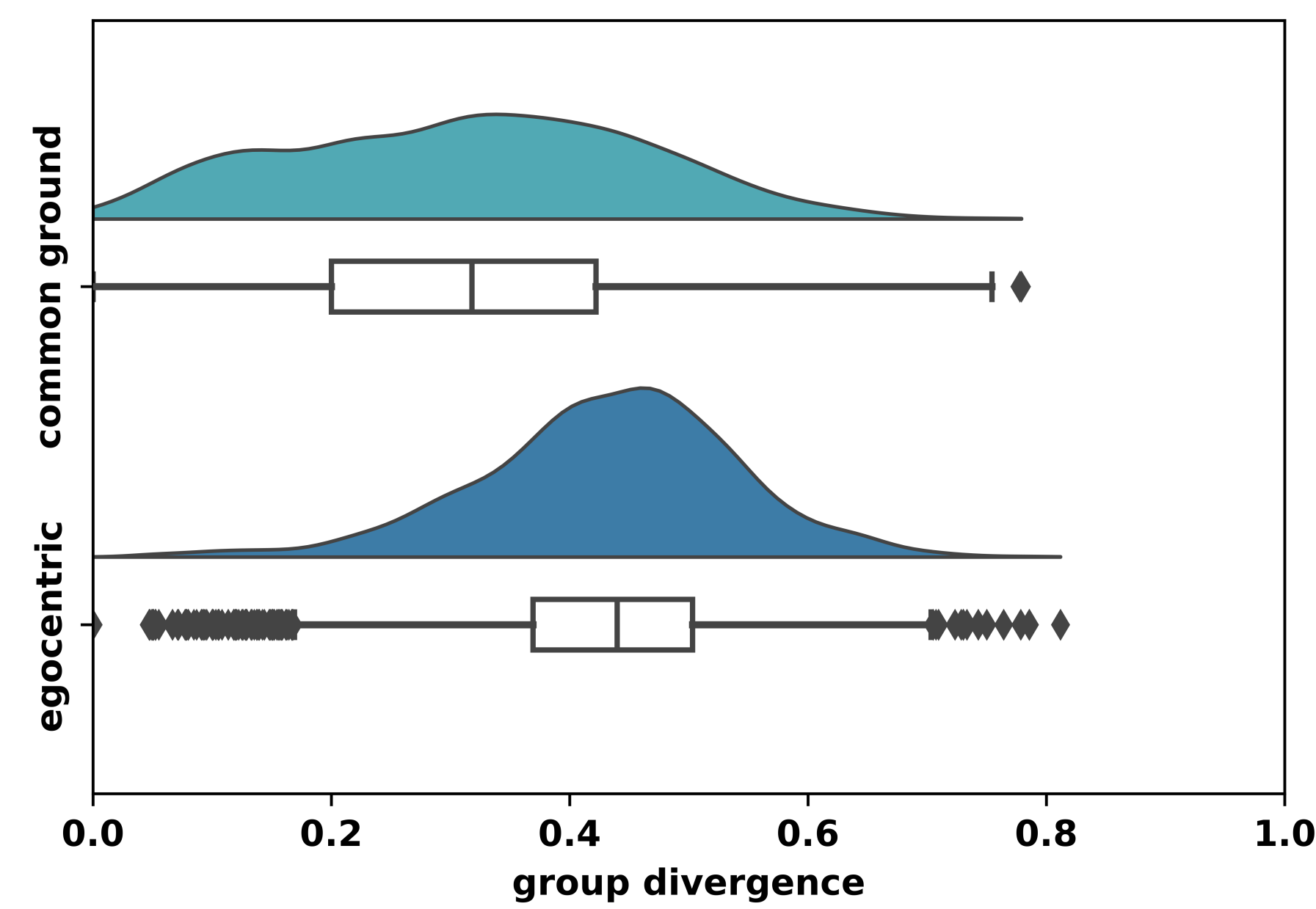
Data from 1,000 runs Details: [8, Ch. 6]

Converging on the truth (2 ag.)



Data from 1,000 runs Details: [8, Ch. 13]

Belief polarisation (12 agents)



Data from 1,000 runs Details: [9]

Discussion & future directions

No uniform epistemic effects: Myside bias does not necessarily lead to negative epistemic outcomes — premise selection matters, at least in artificial agents.

Epistemic vices: Reasoning to support prior beliefs should not necessarily be considered an epistemic vice — but reasoning exclusively from them might be.

Open questions: Can the effects of premise selection be observed in humans? What are, normatively, adequate ways of engaging with the beliefs of others?

References

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